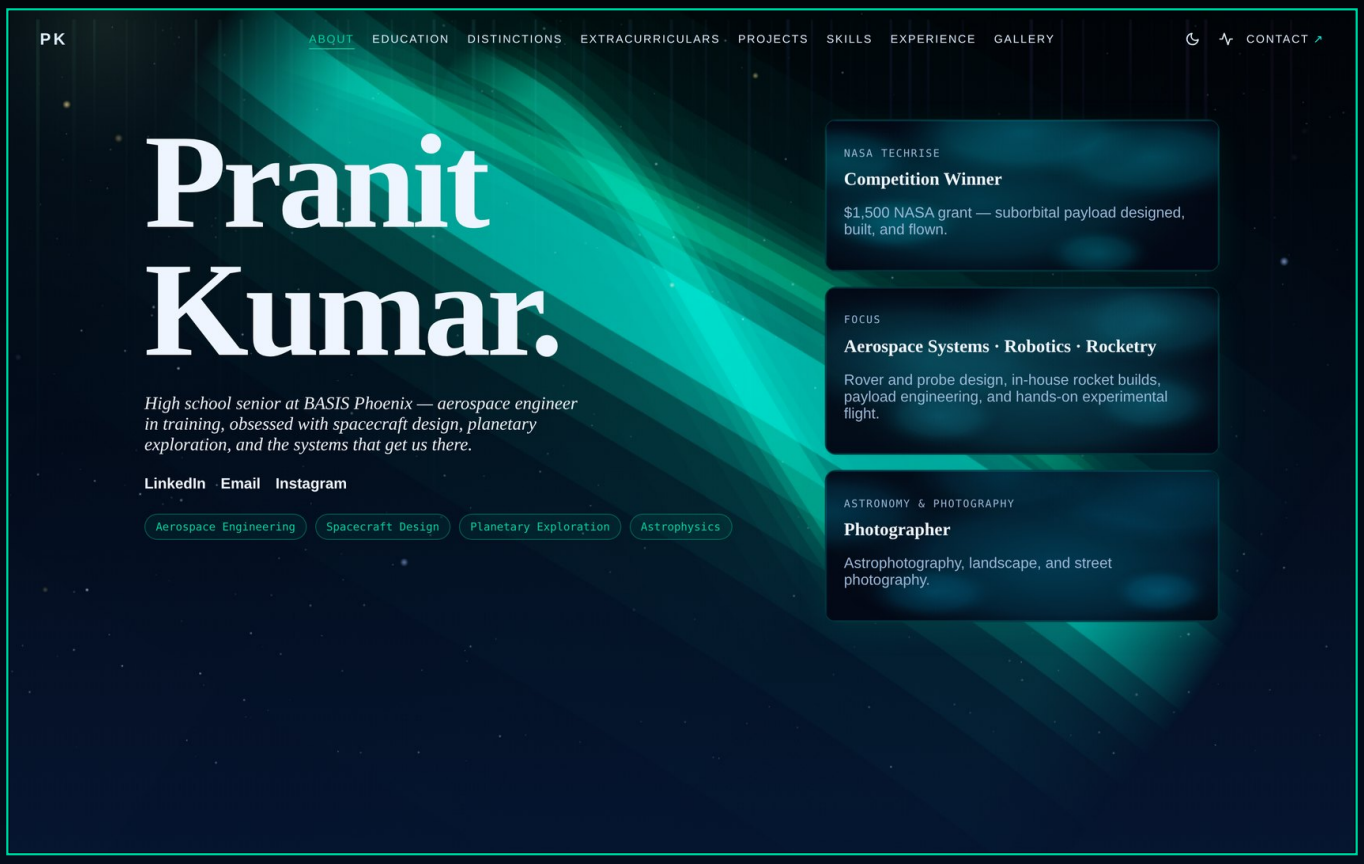
// DESIGN DOCUMENT



pranitkumar.com

A personal site designed like the sky it is about.

Concept · Visual language · Motion · Branding · Marketing · Architecture

VERSION 1.0 · 27 OCTOBER 2026
Prepared for Pranit Kumar
COMPANION TO THE MODULAR REFACTOR OF THE SITE SOURCE
Screenshots: live site, 1440 px desktop and 390 px mobile

// OVERVIEW

The site in one idea

pranitkumar.com is the personal site of a high-school senior who wants to become an aerospace engineer. Its design thesis is simple: **a portfolio about spacecraft and planetary exploration should feel like looking up**. *The page is a sky. In daylight it is soft blue with drifting clouds; after sunset it becomes a night sky with stars and a slowly moving aurora. Content sits on glass cards in that sky, set in a confident editorial serif.*

This document records **why** the site looks and behaves the way it does — concept, audience, visual language, motion, brand voice and marketing — and shows the result with screenshots of the real site. A final part describes the engineering that keeps the design maintainable, including the modular refactor delivered with this document, and lists the issues we found along the way.

PART	CONTENT	PAGES
A. Strategy	01 Concept & narrative · 02 Audience & goals · 03 Information architecture	3–7
B. Visual language	04 Colour · 05 Typography · 06 Layout & components · 07 Day theme · 08 Night theme	8–14
C. Experience	09 Motion & interaction · 10 Responsive design · 11 Time-based personalisation · 12 Secondary pages	15–20
D. Brand & marketing	13 Brand platform & voice · 14 Marketing & distribution	21–22
E. Engineering	15 Architecture & refactor results · 16 Accessibility & risk · 17 Roadmap · Appendix	23–28

Five design pillars

PILLAR	WHAT IT MEANS	WHERE YOU SEE IT
1. The page is a sky	One continuous environment instead of boxed sections; theme changes the whole atmosphere.	Fixed sky canvas, day clouds, night aurora
2. Evidence first	The strongest proof (NASA TechRise win, $1,500 grant) is in the first screen, not buried.	Hero card stack, Distinctions section
3. Editorial confidence	A huge serif name, restrained palette, generous whitespace, mono labels like code comments.	“Pranit Kumar.” at up to 144 px, // SECTION labels
4. Quiet delight	Motion rewards attention but never blocks reading; everything degrades gracefully.	Liquid cursor, theme flash, reduced-motion path
5. Honest engineering	The site is itself a demonstration: solar maths, off-thread rendering, no frameworks.	Time-based theme, worker-rendered aurora

01 / STRATEGY

Concept & narrative

A site is more memorable when it has one metaphor that explains every decision. Here the metaphor is **altitude**. Pranit's work spans the whole vertical range of aerospace: a classroom in Phoenix, a high-altitude balloon payload, model rockets, and the planets that spacecraft are built to reach. The site lets the visitor travel that range by changing the sky itself.

Day = the atmosphere

Daylight is the "Earth" state: a warm blue gradient (#5ba3e0 to #c9e8ff), procedurally laid-out clouds on the sky and a soft cloud behind every card. It is friendly, readable and approachable — the right first impression for an admissions reader at 10 a.m.

Night = space

After sunset the same page becomes the dark side of the planet: near-black navy, a field of twinkling stars and a drifting aurora ribbon, with glowing aurora behind each card. It signals depth and technical craft, and it is the state that makes people screenshot the site.

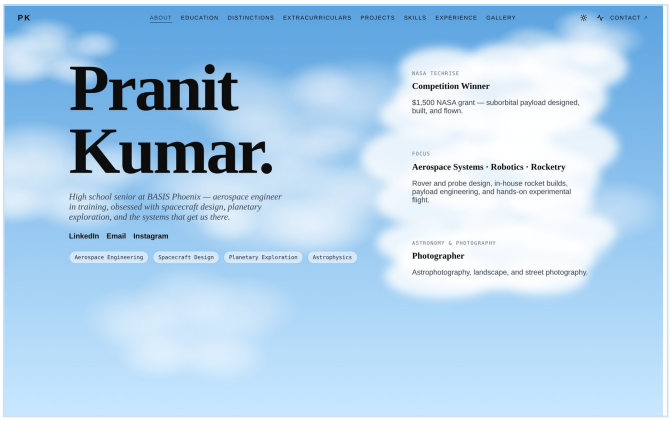


Fig. 1 — Day: the atmosphere. Clouds sit behind the hero and each card.

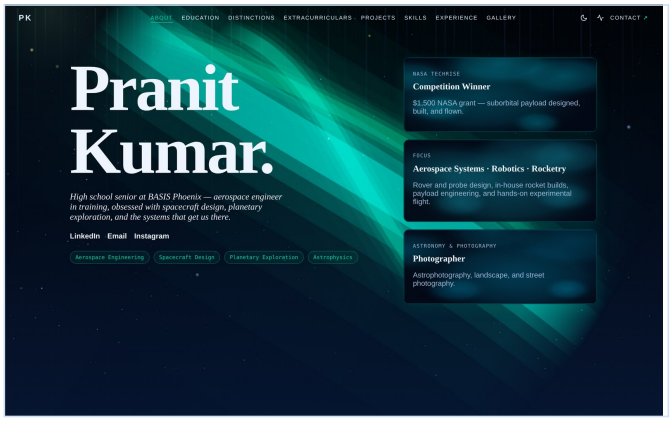


Fig. 2 — Night: space. Aurora ribbon, stars and glowing cards.

> *The visitor never "switches to dark mode." They arrive at whichever sky matches their local time — the page behaves like the planet it is about.*

The narrative arc of the page

Content follows a deliberate story, in the order a reviewer wants to hear it:

- **Who** — name, one-line identity, three focus areas (hero).
- **Foundation** — education and rigour (BASIS Phoenix, AP record, Embry-Riddle course).
- **Proof** — distinctions, extracurricular leadership (two co-founded teams).
- **Work** — projects with links to real artefacts (experiment PDF, PCB paper).
- **Range** — skills, experience, and a photography gallery that shows the human behind the engineering.

- **Next step** — a plain contact list. No forms, no friction.

02 / STRATEGY

Audience & goals

The site has a small, high-intent audience. Each segment arrives with one question and about two minutes of attention.

SEGMENT	THEIR QUESTION	WHAT THEY NEED TO SEE	SITE ANSWER
College admissions readers	Is this applicant exceptional and genuine?	Rigour, leadership, a distinctive project, voice	Education numbers, co-founder roles, NASA win, writing tone
Program / internship reviewers	Can this student do real engineering?	Artefacts, tools, results	Experiment PDF, I2C PCB paper, skills grid
Teachers & mentors	Is this person growing and credible?	Breadth, service, consistency	Volunteering, instruction experience, honours
Peers & collaborators	Is this someone I want to build with?	Personality, craft, links	Gallery, @pranit.pics, playful details
Curious visitors	What is this?	A reason to stay	The sky itself, the cursor, the theme choice

Goals and success measures

GOAL	HOW WE KNOW IT WORKS
Make a credible first impression in under 10 seconds	Hero shows name, identity line and NASA proof without scrolling on a 1440×900 screen (Fig. 1).
Reward a deeper read	Every claim has a nearby artefact or number; sections are scannable with // labels and card titles.
Be memorable without being gimmicky	Atmosphere is ambient (sky, glow), interaction is optional (cursor, theme popup); content stays 100% legible.
Work for every visitor	Mobile layout, reduced-motion path, no-JS day palette, graceful fallback when workers are unavailable.
Be maintainable by one person	No framework, no build step; modular code and tokens documented in the README and in Part E.

03 / STRATEGY

Information architecture

The site is deliberately shallow: **one long home page, one gallery, one 404**, plus two standalone utilities. A single scroll keeps the narrative intact; the navigation is a scroll-spy that always shows where you are.

PAGE	PURPOSE	NOTES
index.html	The portfolio narrative (9 sections)	About · Education · Distinctions · Extracurriculars · Projects · Skills · Experience · Gallery call-to-action · Contact
gallery.html	Photography and project images	Driven by gallery/manifest.json; broken images disappear instead of leaving holes
404.html	Lost-in-space error page	5-second redirect with a progress bar, brand voice, same sky
fun.html	Playful standalone page	Self-contained fake-opportunity popups; intentionally separate from the real site's claims
japanesestrokeordertool.html	Standalone utility	Japanese stroke-order viewer; independent paper-white design

Labelling system

Section titles are conversational sentences ending in a period — Where I've studied. What I do. Things I've built. — paired with a mono label that reads like a code comment (`// EDUCATION`). The pairing says two things at once: a human voice, and an engineer's precision.

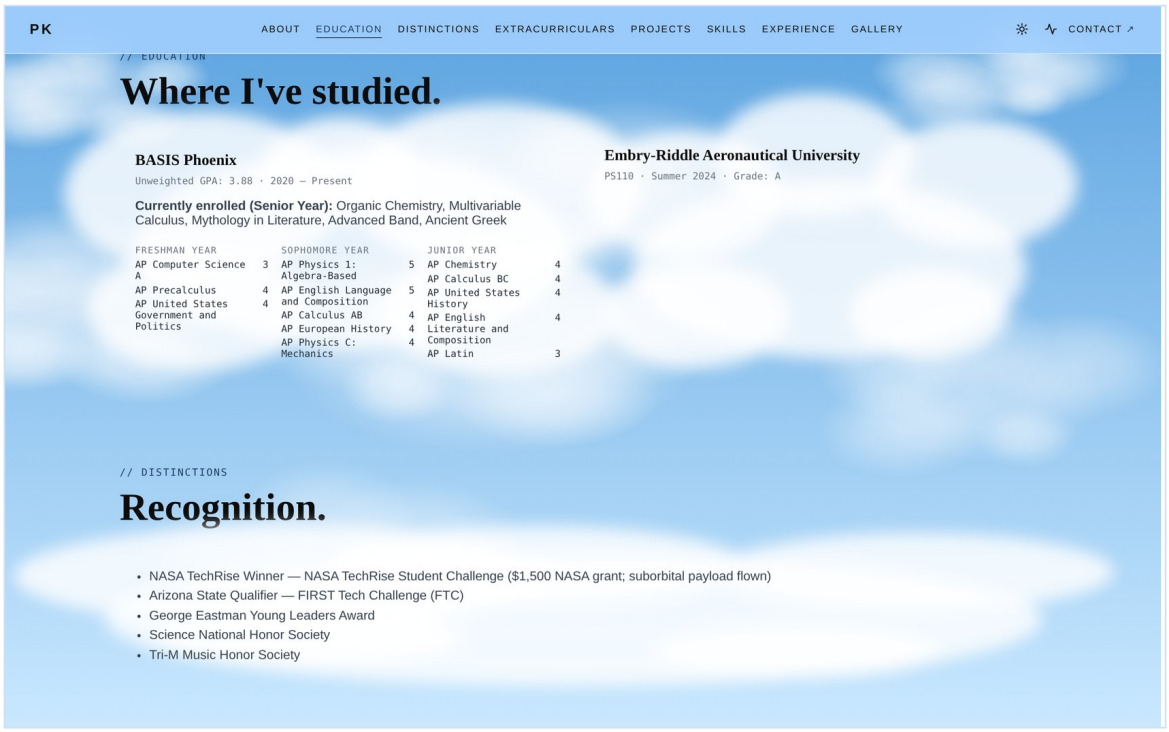


Fig. 3 — Education section: mono label, sentence-style heading, glass cards, AP scores as a compact three-column grid (day).

Content priorities

- **Numbers over adjectives.** GPA, AP scores, grant amount and dates appear as facts; the voice stays modest.
- **Links to artefacts.** Projects carry pill links (“Experiment PDF”, “Paper”) styled as small mono buttons.
- **Contact is a list, not a form.** Five labelled lines (email, LinkedIn, Instagram, photography, website) — zero infrastructure and zero spam surface.



Fig. 4a — The entire home page at 1440 px (day).



Fig. 4b — The same page at night: one continuous sky, nine sections.

04 / VISUAL LANGUAGE

Colour system

All colours are tokens in one stylesheet (css/theme.css). A theme is simply a second set of values for the same token names, so every component works in both skies without extra rules. The day palette is cool, bright and low-saturation; the night palette is nearly black with two luminous accents (aurora green, electric cyan).

Day tokens



SKY --sky-top #5ba3e0 to --sky-bottom #c9e8ff

#0D0D0D	#374151	#6B7280	#1E3A5F
Ink	Slate	Grey	Harbour navy
--text-primary	--text-secondary	--text-muted	--accent, --text-label

#5BA3E0	#C9E8FF	#E8F2FC	#1F2937
Sky top	Sky bottom	Mist	Chip text
--sky-top	--sky-bottom	card glass at 82%	--chip-text

Night tokens



SKY --sky-top #010407 to --sky-bottom #061530

#EEF5FF	#9AB8D8	#6A8AAA	#00D2A0
Moonlight	Blue grey	Slate blue	Aurora green
--text-primary	--text-secondary	--text-muted	--text-label, chips

#03FCF0	#061530	#010407	#040816
Electric cyan	Deep space	Void	Card glass
--accent	--sky-bottom, overlay	--sky-top	card glass at 85%

Colour rules

- **One accent per theme.** Navy by day, cyan by night. Accents mark interaction and emphasis only — never body text.
- **Glass, not boxes.** Cards are translucent surfaces (82–85% opaque) with a 1 px light border and a soft coloured shadow; the sky always shows through.
- **Aurora hues are rendered, not tokenised.** The night animation sweeps green (hue 148) to violet within the canvas; the CSS palette stays calm so text remains legible.
- **No hard-coded colours in components.** Components reference variables; the only literals are the white icon on active buttons and pure-black theme-switch flash.



Fig. 5a — Day navigation, ink text on sky.



Fig. 5b — Night navigation, aurora-green active link.

05 / VISUAL LANGUAGE

Typography

Three families, three jobs. The contrast between a high-contrast display serif, a geometric sans and a monospace is the site's signature: **editorial, modern, technical**.

FAMILY	ROLE	WEIGHTS USED	SPECIMEN
Playfair Display	Name, section headings, card titles, taglines (italic)	400 · 400i · 700 · 900	Pranit Kumar.
Space Grotesk	Body copy, navigation, buttons	400 · 500 · 700 · 800	Aerospace Engineering
IBM Plex Mono	Section labels, metadata, chips, footer, dates	400 · 500	// PROJECTS · Sep 2024

Type scale

LEVEL	SIZE (CSS)	NOTES
Hero name (h1)	clamp(4rem, 10vw, 9rem) · weight 900 · line-height .95 · tracking −.03em	64–144 px. The most expressive element: tight leading, heavy weight, terminating period.
Section title (h2)	clamp(2rem, 4vw, 3rem)	Sentence case with a full stop: a human sentence.
Card title (h3)	clamp(1rem, 2.5vw, 1.2rem)	Playfair 700; always sits under a mono label.
Tagline	clamp(1.1rem, 2.2vw, 1.2rem) italic	One 480 px measure; the emotional sentence of the page.
Body	~0.95–1.1 rem Space Grotesk	Slate (day) / blue grey (night); line-height about 1.5.
Mono label	.7–.78 rem · letter-spacing .08–.12em · uppercase	Always muted or label-coloured; never a heading.

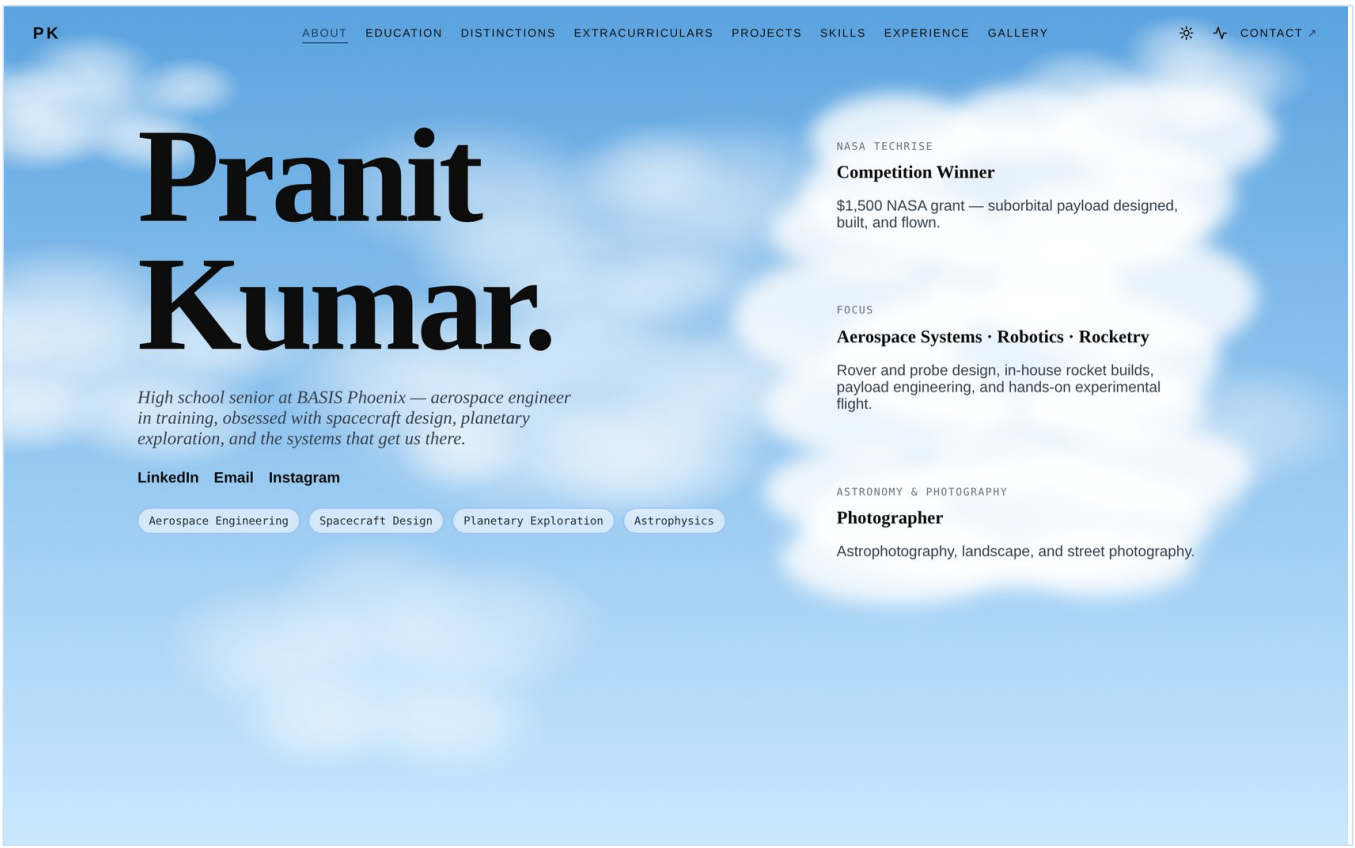


Fig. 6 — Type in context: heavy serif name, italic tagline, bold sans links, mono chips (day, 1440 px).

Why this works

- The **serif/sans/mono triangle** maps to the three audiences: editorial (admissions), clean (general), technical (engineers).
- **Fluid sizing** with clamp() means no breakpoint jumps; the name scales from phone to ultrawide.
- **Mono labels at small sizes** add texture without adding noise — they behave like captions on a technical drawing.

06 / VISUAL LANGUAGE

Layout & components

The grid is intentionally simple: a centred 1200 px container with 1.5 rem gutters; everything else is CSS Grid with one-to-three columns. Vertical rhythm comes from a 3.5 rem section padding that tightens to 2 rem on phones.

COMPONENT	SPEC	NOTES
Hero	Grid 1.65fr / 1fr · 2 rem gap · min-height about one viewport	Left: identity. Right: three evidence cards.
Card	12 px radius · 1.1 × 1.2 rem padding · glass surface	Background layer is painted by JS (clouds/aurora); content is lifted above it.
Chip	999 px radius · mono .74 rem · tinted fill + border	Four focus keywords under the hero.
Link pill	999 px radius · 1 px ink border · mono .76 rem	Hover: accent border (day). Used for artefact links.
Education grid	3 equal columns (1 on phone)	AP scores: course left, score right, grouped by year.
Skills grid	2 columns, each row 170 px label + text	Rows are cards; collapse to stacked on phone.
Contact list	130 px label + value, hairline separators	Max 680 px; no form.

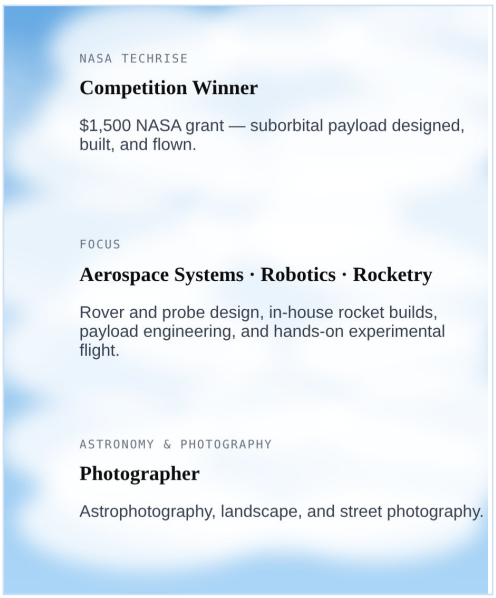


Fig. 7a — Hero cards in day: cloud behind the text, no hard border.

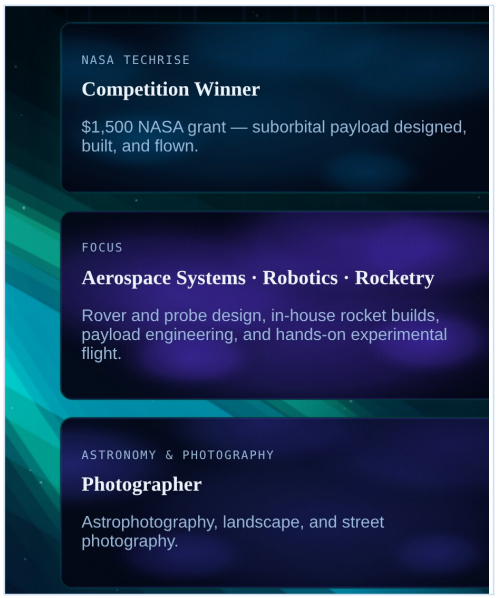


Fig. 7b — Same cards at night: dark glass, 1 px aurora border, glowing canvas.



Fig. 8a — Chips and links, day.



Fig. 8b — Chips and links, night.

Layering (z-order)

The page is built from strict layers so effects never intercept clicks and controls never hide behind decoration:



07 / VISUAL LANGUAGE

Day theme: the atmosphere

The day sky is a layered illusion built with plain canvas and CSS — no images. Two systems work together:

1. Sky clouds (canvas)

Seven clouds are placed on a loose 3-column grid with random jitter so they never form a pattern. Each cloud is six overlapping radial-gradient ellipses with fading edges. The layout is cached per tab, so moving between pages does not reshuffle the sky; a hard refresh deliberately generates a new one.

2. Card clouds (DOM)

Every card gets a blurred, slightly larger cloud made from six absolutely-positioned ellipses (10 px blur, 36 × 28 px bleed beyond the card). Each cloud bobs by 6 px on its own 4.5–7.2 s loop, with a random delay, so neighbouring cards never move in lockstep.

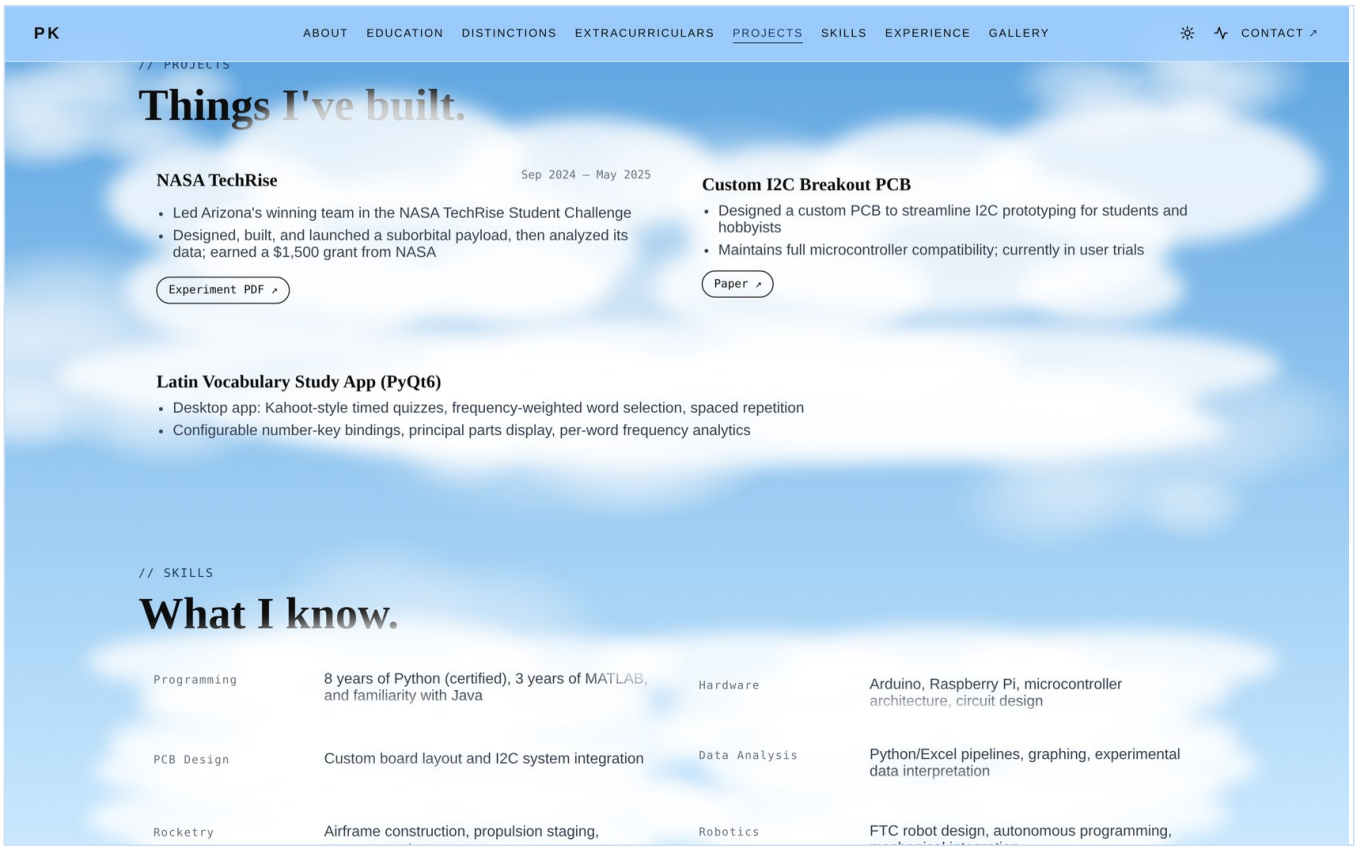


Fig. 9 — Projects in day: cloud-backed cards over the sky; pill links in mono.

Design intent

- Clouds are **soft and slightly transparent**: text contrast stays high (primary text 17.6:1) while the surface still reads as atmosphere.
- **Motion is tiny.** A 6 px float is perceived as “alive”, not “animated”.
- The day theme is the **default for no-JS visitors** (tokens live in CSS), so the page is complete even if scripts never run.

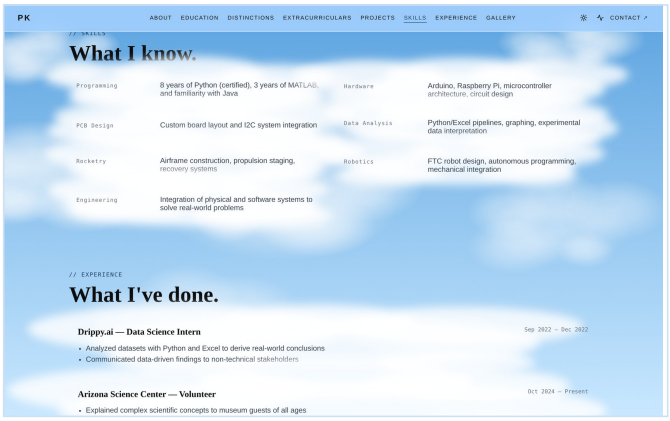


Fig. 10a — Skills: label/value rows as cards (day).

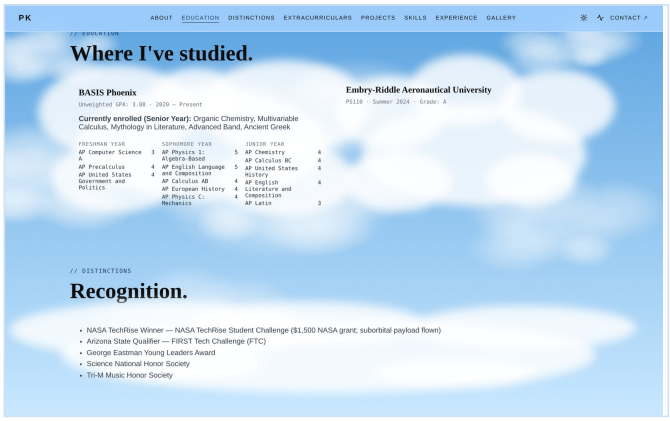


Fig. 10b — Education cards and AP grid (day).

08 / VISUAL LANGUAGE

Night theme: the aurora

The night sky is the site's showpiece and its most expensive effect, so it is also its most carefully engineered. It is rendered on **canvas in a Web Worker**, leaving the main thread free for scrolling and input.

LAYER	DESCRIPTION	COST CONTROL
Stars	400 faint stars (200 on phones) with individual twinkle speeds, plus 20 larger warm/cool "special" stars with soft glow	Count halves on mobile
Rays	Up to 65 thin vertical aurora curtains with a green-to-violet hue sweep, each swaying slowly	30 on phones; simpler blending in performance mode
Ribbons	One bright hero ribbon plus two accents, each drawn as 9 stacked bands of falling opacity so edges feather like real aurora	Performance mode: 3 bands, fewer segments
Focal glow	A cyan-white bloom near the top-left that pulls the eye toward the name	Single gradient
Card aurora	A per-card canvas: six jittered blobs that slowly shift colour between two palette hues over dark glass	24 fps, only when within 120 px of the viewport

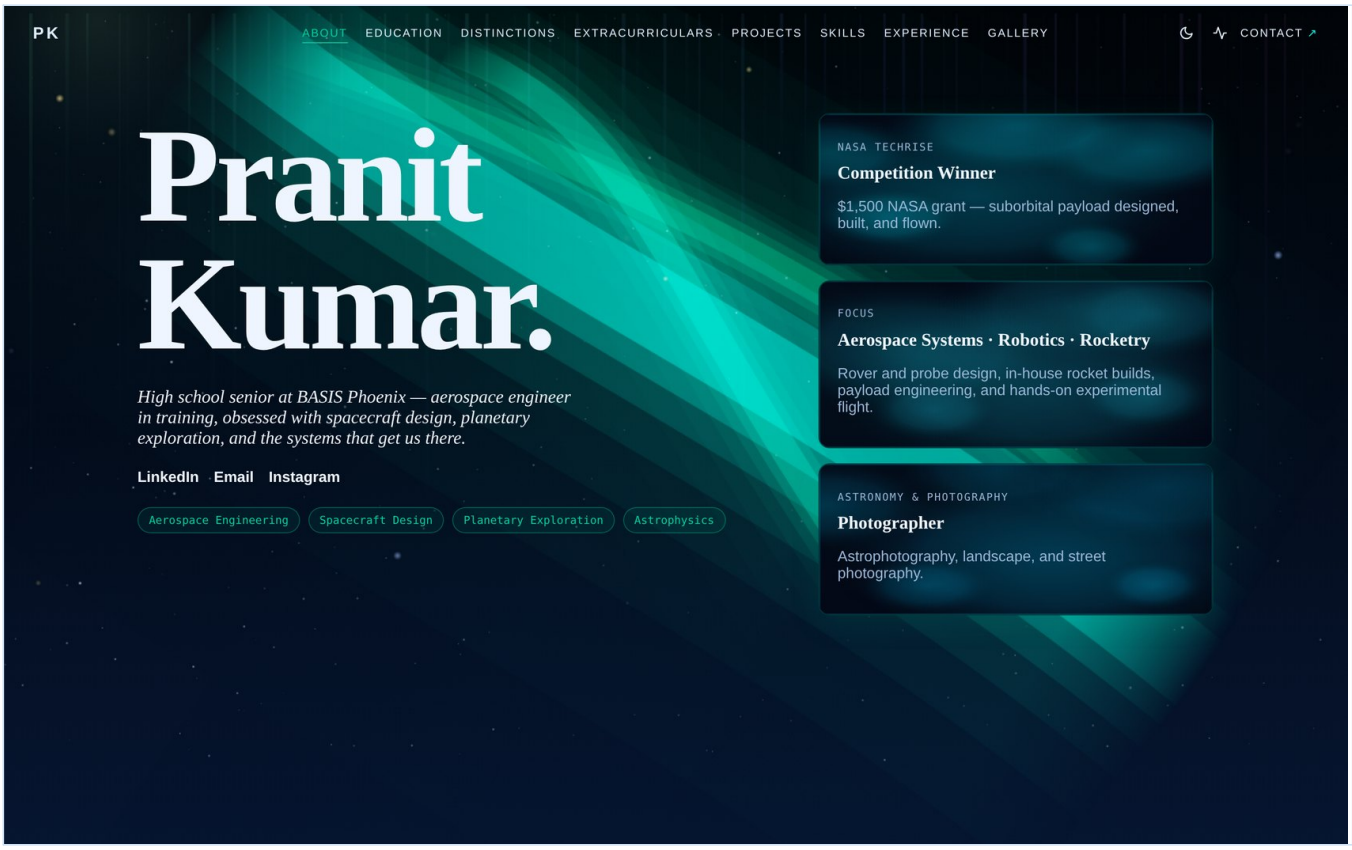


Fig. 11 — Night hero: hero ribbon sweeping from top-left, rays above, stars across the whole field.

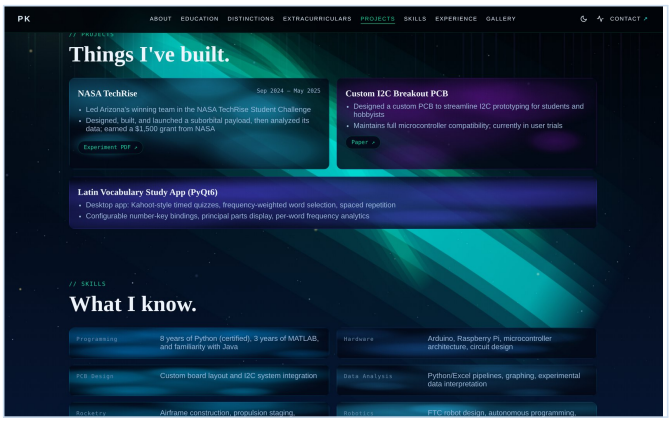


Fig. 12a — Projects at night: aurora glow behind each card.

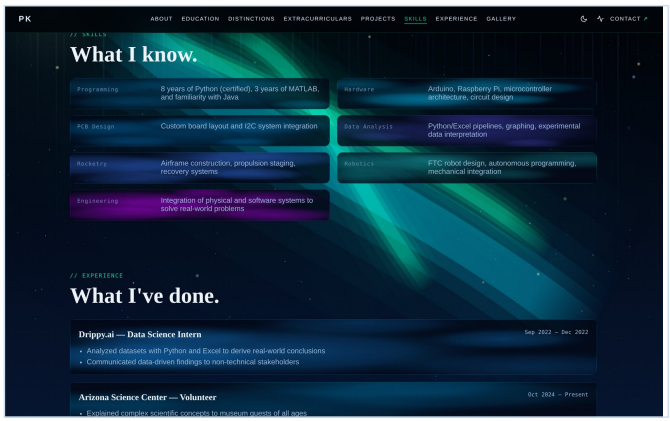


Fig. 12b — Skills at night: every row card has its own glow.

Legibility decisions

- Card surfaces are 85% opaque deep navy, so aurora colour tints the card but never competes with text (primary text 18.0:1).
- Paragraphs at night use blue grey (#9ab8d8, 9.6:1); labels and chips use aurora green (10.1:1).
- **Performance mode** is offered to the visitor in the theme popup and navigation: it removes expensive glow passes so low-power devices stay smooth.
- Reduced-motion visitors receive a single still frame of the same sky — the beauty without the movement.

09 / EXPERIENCE

Motion & interaction

Motion in this site is **ambient or responsive — never required.** Content is readable with every animation off.

The liquid cursor

On devices with a fine pointer, the system cursor is replaced by a small orb that trails the mouse with easing. It has three states, which together signal how the page responds:



Fig. 13a — Moving: compact solid orb (7 px).

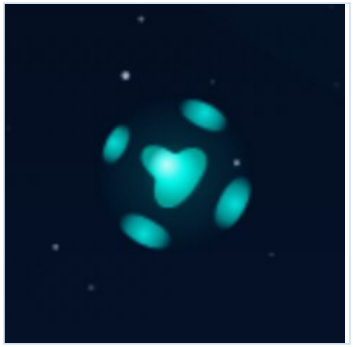


Fig. 13b — Idle 1.2 s: grows, wobbles, sheds orbiting blobs.



Fig. 13c — Over a link or button: 28 px bloom.

MOMENT	MOTION	PURPOSE
Pointer moves	Orb eases toward the pointer (18% per frame); radius 7 px	Precision while working
Pointer rests 1.2 s	Effect fades in slowly, orb grows to 10 px, four blobs orbit	Delight and a sense of life

MOMENT	MOTION	PURPOSE
Pointer over clickable	Orb expands to 28 px, blobs orbit faster, a bloom pulse fires	Affordance: “this is interactive”
Theme switch	280 ms fade to black, theme swaps while hidden, 80 ms hold, fade out	Masks the canvas rebuild so nothing flickers
Mobile menu	Overlay fades/slides in; links stagger 40 ms apart top-to-bottom (reversed on close)	Orientation and polish
Scrolling	Topbar gains blur and border after 8 px; scroll-spy highlights the current section	Always know where you are
Cards	Day: 6 px float, 4.5–7.2 s; Night: 24 fps aurora while on screen	Ambient life at low cost

Accessibility of motion

- **prefers-reduced-motion** disables transitions/animations, uses a still sky and does not activate the custom cursor.
- **Touch devices** keep the native cursor; the orb is a mouse-only refinement.
- Everything the orb does is cosmetic — hover states on the real elements remain (colour, border, underline).

10 / EXPERIENCE

Responsive design

The site has one deliberate breakpoint at 768 px. Below it the hero stacks, grids collapse to a single column, desktop navigation is replaced by a full-screen menu, and the theme controls move into a compact topbar button beside the hamburger.

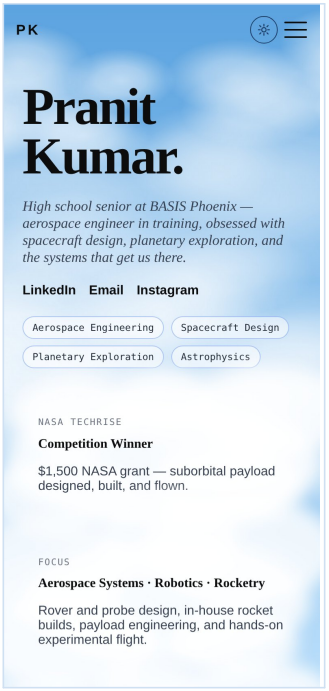


Fig. 14a — Phone, day: stacked hero, name scales to 64 px.

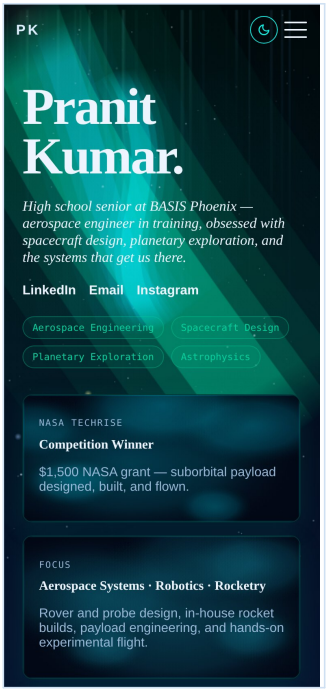


Fig. 14b — Phone, night: lighter star and ray counts.



Fig. 14c — Full-screen menu, staggered links.



Fig. 14d — Menu at night.

AREA	DESKTOP	PHONE (768 PX AND BELOW)
Navigation	Eight centred text links, Contact on the right	Hamburger opens a full-screen overlay generated from the same link list
Hero	Two columns (1.65 : 1)	One column, auto height
Grids	2–3 columns	1 column; skills rows stack label over text
Theme controls	Two icon buttons in the topbar rail	One round button left of the hamburger; popup becomes a centred modal
Night effects	400 stars · 65 rays	200 stars · 30 rays · 3 special stars
Background	Gradient on body, fixed	Gradient pinned with a fixed layer (iOS ignores fixed backgrounds)

11 / EXPERIENCE

Time-based personalisation

First-time visitors are invited — after 0.9 s — to **choose their experience**: Light, Dark or Time-based, plus a performance switch. The choice is remembered. In Time-based mode the site decides by real daylight, not by the clock.

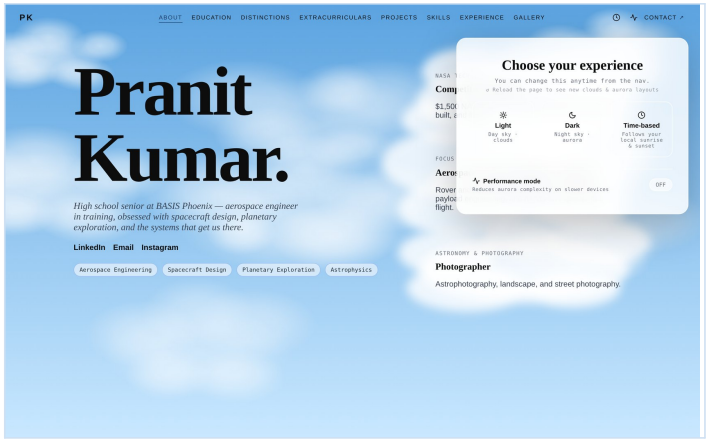


Fig. 15a — Desktop: popup anchored top-right over the live day sky.

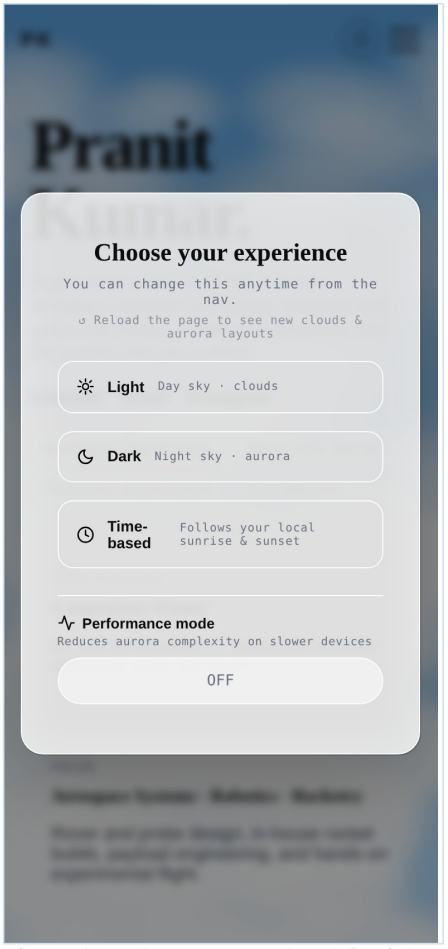


Fig. 15b — Phone: centred modal with a dimmed, blurred backdrop.

How “Time-based” works

No location permission is requested. The browser reports its IANA time zone (e.g. America/Phoenix); a small table maps every zone to a representative latitude and longitude. From those two numbers the site computes sunrise and sunset:

- **Solar declination** for today’s date: declination = 23.45° × sin(2 × pi × (day − 81) / 365).
- **Day length** from the hour-angle formula: H = acos(−tan(latitude) × tan(declination)).
- **Solar noon** is shifted by how far the longitude sits from the zone’s standard meridian (4 minutes per degree).
- Polar day and night are handled by clamping; a missing zone falls back to a latitude estimated from the UTC offset.

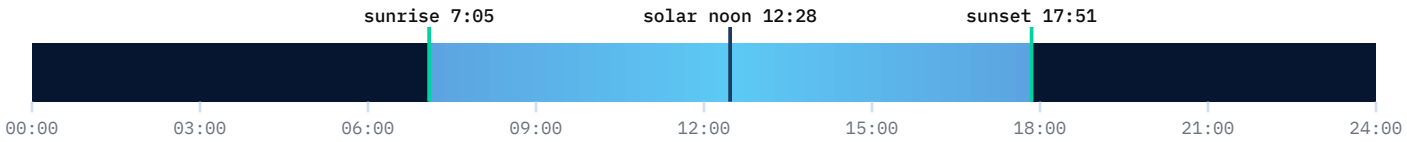


Fig. 16 — Computed daylight for Phoenix on 27 Oct 2026 (latitude 33.45°, longitude −112.07°): the site is "day" between about 07:05 and 17:50 and "night" the rest of the time. Solar noon is 12:28, not 12:00, because Phoenix sits west of its time-zone meridian.

> *A recruiter in Phoenix at noon and a college counsellor in Singapore at midnight each meet a sky that matches their own world — a small act of hospitality, and a quiet proof of applied maths.*

The theme is applied **before first paint** by a tiny blocking script, so there is never a flash of the wrong sky. While a tab stays open in Time-based mode, the site re-checks every minute and transitions at sunrise or sunset.

12 / EXPERIENCE

Secondary pages & brand moments

Every page is an opportunity to repeat the brand. The secondary pages use the same sky, type and glass language, and each adds a small moment of personality.

Gallery

A responsive grid of photographs (auto-fit columns of at least 260 px, 340 px tall crops) driven by a JSON manifest. Hidden-until-loaded figures and automatic removal of broken images mean the grid never shows empty frames. Captions are supported; alt text currently falls back to the file name (see Roadmap).



Fig. 17a — Gallery, day.

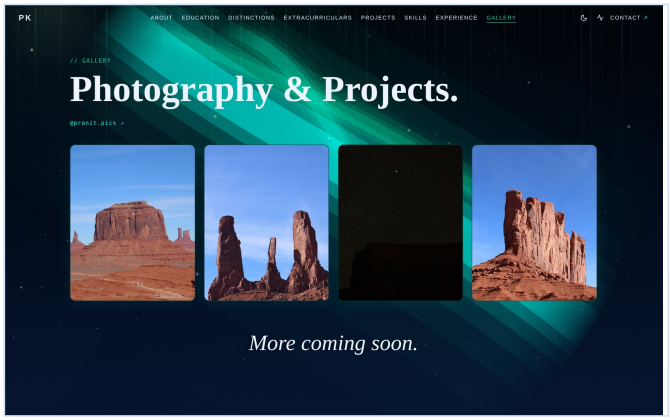


Fig. 17b — Gallery, night.

404: "This page drifted off course."

A missing page is reframed as a space-flight mishap, with an honest redirect: a five-second progress bar and a clear "Return home" button. The error is a brand moment, not a dead end.

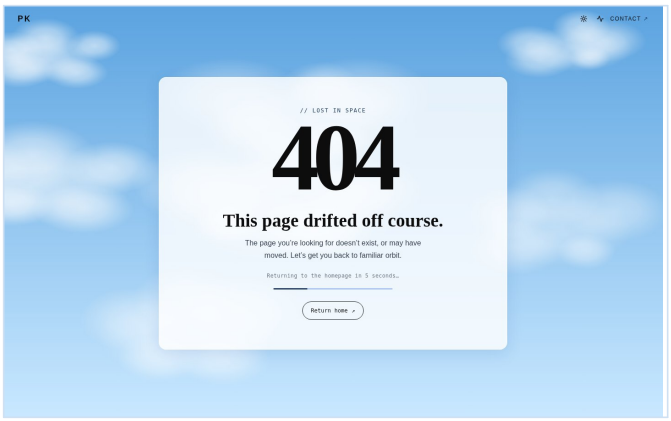


Fig. 18a — 404 in day.

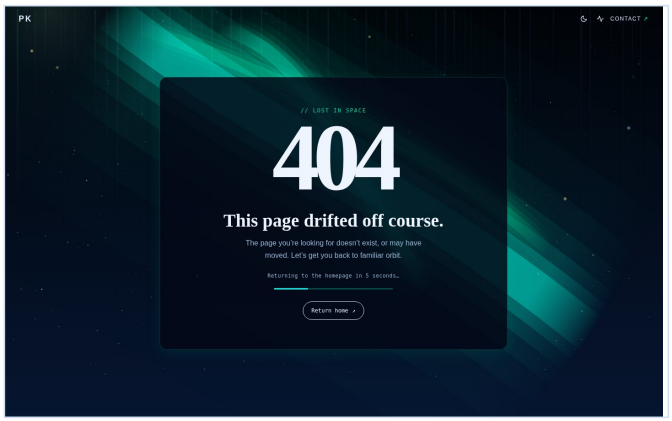


Fig. 18b — 404 at night.

Fun page and utility

fun.html is a deliberate joke: a stream of draggable fake "opportunity" popups (NASA, MIT, SpaceX …) that all lead to a classic prank link. It shows self-awareness and humour; it is kept fully separate from the real site so its claims can never be mistaken for real ones. **The stroke-order tool** is a standalone Japanese-writing utility in a paper-white, red-accent style — evidence of curiosity beyond aerospace.

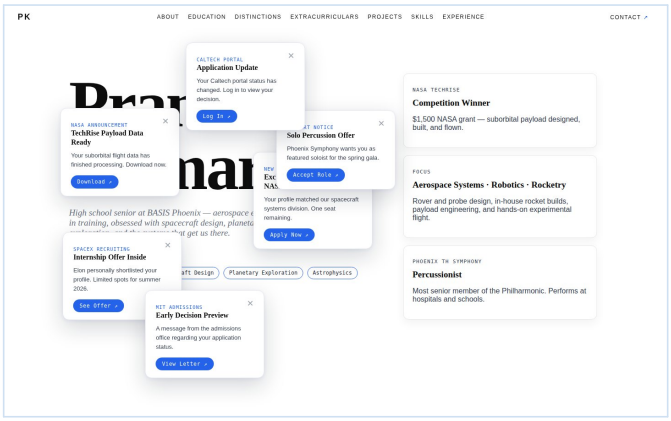


Fig. 19a — fun.html: draggable fake popups over a clean white page.



Fig. 19b — Stroke-order tool: its own visual identity.

13 / BRAND

Brand platform & voice

Positioning

> **For people deciding whether to bet on a young engineer,** *pranitkumar.com shows a student who builds real hardware, leads teams and communicates clearly — presented with the polish and curiosity of someone who loves the sky.*

Brand personality

TRAIT	EXPRESSED THROUGH	AVOID
Curious	Sky metaphor, mono “code comment” labels, the stroke-order tool	Gimmicks that distract from content
Precise	Numbers, dates, links to artefacts, tidy grids	Vague superlatives (“passionate”, “world-class”)
Warm	Sentence-style headings, a tagline in italic, volunteer and music work	Cold corporate phrasing
Playful	Liquid cursor, 404 copy, the prank page	Humour inside real claims or applications
Ambitious	NASA, spacecraft, planetary exploration	Overstating roles or outcomes

Voice & tone guide

PRINCIPLE	DO	DON’T
Lead with the fact	“$1,500 NASA grant — suborbital payload designed, built, and flown.”	“Incredibly proud to have been honoured with …”
Use active, verb-first bullets	“Designed a custom PCB to streamline I2C prototyping.”	“Responsible for the design of …”
Headings sound like speech	“Where I’ve studied.” “Things I’ve built.”	“Academic Background”
Keep it short	One-line taglines, two-bullet projects	Paragraphs of biography
Let the work speak	Link the experiment PDF and paper	Unsupported claims

Identity assets

- **Monogram “PK”** — Space Grotesk 800, tracked +.16 em, top-left of every page, links home.
- **Name lock-up** — “Pranit Kumar.” in Playfair 900 with a closing full stop: a statement, not a label.

- **Handles** — @pranit.kumar1 (personal) and @pranit.pics (photography): two audiences, one site; the gallery bridges them.
- **Domain** — pranitkumar.com: short, literal, trustworthy. Email on the same domain reinforces credibility.
- **Colour memory** — blue day sky and green-cyan aurora are the two brand colours; keep print and social assets within them.

Related collateral already in the repository: a one-page **FTC engineering portfolio cover** (team Firebytes, "Decode" 2026) and an **I2C breakout board paper**. Both can be restyled with the same palette and type to feel like one family.

14 / BRAND

Marketing & distribution

A personal site markets a person, not a product. The objective is **qualified attention**: the right ten people reading the site closely beats ten thousand drive-by visits.

Funnel



CHANNEL	ROLE	ACTION	STATUS
Application materials	Highest-intent traffic	Put pranitkumar.com in every résumé header, activity list and email signature; link directly to #projects or #distinctions when relevant	Do
LinkedIn	Professional discovery	Website in profile; post project milestones (TechRise, PCB trials) with a screenshot of the night hero	Do
Instagram @pranit.pics	Photography audience	Bio link to gallery; use the aurora/sky look in story templates	Do
Engineering portfolio PDF	FTC judges and mentors	Add a QR code and URL to the portfolio cover page	Do
Search	Name searches	Meta description exists on the home page; add titles/descriptions for gallery and structured data (Person)	Improve
Link previews	Messages, social, Slack	**No Open Graph/Twitter tags today** — shares show a plain link. Add og:title, og:description and a 1200×630 night-hero image	Fix first

Messaging pillars

- **Proof:** "NASA TechRise winner — suborbital payload designed, built and flown."
- **Leadership:** "Co-founded the Aerospace Engineering Club and the FTC team."
- **Range:** "Rockets, robots, PCBs, a Python curriculum for kids, and a photographer's eye."

Measurement (privacy-respecting)

The site currently has no analytics. For a personal site, a cookie-free tool (or simple server logs) is enough. Track: visits by referrer (which application channel works), scroll depth to #projects, clicks on Experiment PDF / Paper / Email, and gallery views. Review monthly during application season; otherwise quarterly.

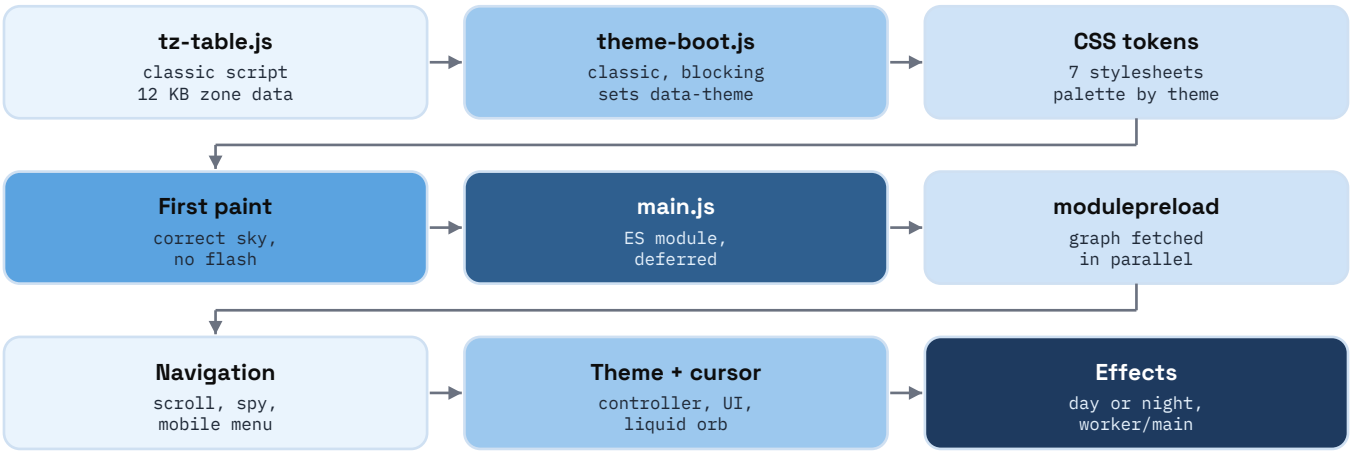
> *Keep marketing quiet: one canonical link, a strong preview image, and a first screen that proves the claim. The design does the persuading.*

15 / ENGINEERING

Architecture & refactor results

The design above is only sustainable if its code is easy to change. The delivered refactor keeps the "no framework, no build step" rule but replaces five loosely coupled global scripts with a small modular system. **Visual and behavioural output was verified identical** (see below).

Boot sequence



Module map

FOLDER	RESPONSIBILITY	KEY FILES
css/	Tokens, base, effects hooks, nav, components, controls, night overrides, per-page	theme.css · night.css
js/boot, js/data	Pre-paint theme kernel and time-zone table (classic, must block)	theme-boot.js · tz-table.js
js/core	Safe storage, random, motion preference, DOM helpers, per-tab seeds	session-seed.js
js/theme	State, controller (events), UI (buttons, popup), icons	controller.js · ui.js
js/nav · js/cursor	Scroll state, scroll-spy, mobile menu; liquid cursor and its tuning	mobile-menu.js · orb.js
js/effects	Orchestrator, day clouds, night DOM side, card decoration, rendering backends	night-backends.js
js/night	Pure canvas renderer, generators, config, worker entry	renderer.js · night-worker.js

One renderer, two homes

The original site had two independent implementations of the night drawing code (worker and main-thread fallback) that had to be edited together. They are now **one class** (NightRenderer) with no DOM access; the worker backend feeds it OffscreenCanvases, the fallback backend feeds it regular canvases. Output is identical everywhere, and performance mode now works in both.

MEASURE	BEFORE	AFTER
JavaScript files / owners of window globals	6 files · 10 global assignments	30 modules, one entry · 1 global (window.PKTheme)
Stylesheets	1 file, 679 lines, 28 !important	7 core files, 535 lines (+ 2 per-page files), 7 !important (all in a11y/cursor rules)
Inline style attributes and JS style strings	50	0 in markup; geometry only set from JS
Render-blocking JavaScript in <head>	45.0 KB (10.8 KB gzip)	15.2 KB (6.5 KB gzip); zone table reduced 23 to 12 KB
Duplicate drawing code (worker vs fallback)	~150 lines mirrored	0 (single renderer)
Mobile menu markup	Two hand-maintained nav lists per page	Generated from the desktop nav; staggers via CSS variables
Comment lines in JS	257	324 (every module has a header contract)

16 / ENGINEERING

Verification, accessibility & risk

How the refactor was verified

- **Pixel comparison** of 21 screenshots (desktop and phone, day and night, menu open, first-visit popup, gallery, 404, fun page) with seeded randomness: 20 identical; one full-page capture differed by 47 anti-aliasing pixels (0.001%).
- **Computed-style comparison** of ~50 selectors in four theme/viewport states plus hover states: identical except for deliberate changes.
- **Interaction probes** for theme choice, performance toggle, scroll-spy, mobile menu, malformed gallery manifests, blocked storage and the no-Worker fallback: identical, with the intentional differences listed in the README.

Measured contrast

THEME	PAIR	RATIO	WCAG AA (4.5:1 text)
Day	Primary text on card	17.6 : 1	Pass
Day	Secondary text on card	9.3 : 1	Pass
Day	Muted text on card (labels, metadata)	4.4 : 1	**Just below** — darken #6b7280 slightly
Day	Accent/label #1e3a5f on top of sky	4.3 : 1	**Just below** for small text
Day	Navigation ink on sky top	7.2 : 1	Pass
Night	Primary text on card	18.0 : 1	Pass
Night	Secondary text on card	9.6 : 1	Pass
Night	Muted text on sky bottom	5.0 : 1	Pass
Night	Aurora-green label on card	10.1 : 1	Pass
Night	White icon on active button (cyan)	1.3 : 1	**Fail** — use dark icon on cyan

Findings and recommended fixes

#	FINDING	IMPACT	FIX
1	Active theme/performance buttons use white on cyan at night (1.3:1)	Icon nearly invisible when active	Set active icon colour to var(--sky-bottom) at night
2	No visible :focus-visible styles on links/buttons	Keyboard users lose their place	Add a 2 px accent outline with offset
3	Theme popup lacks role="dialog", aria-modal, focus trap and Esc to close	Screen-reader and keyboard friction	Add dialog semantics and focus management
4	Gallery images use the file name as alt text	Meaningless for screen readers	Add an "alt" field to the manifest
5	Muted text at 4.4:1 and navy accent at 4.3:1 in day	Marginal for small text	Darken tokens by ~8%
6	In night, cascade makes card labels secondary-coloured instead of aurora green	Design intent unclear	Decide; document or restore green labels
7	On the 404 page at night, home-link hover turns cyan on near-white	Low contrast on hover	Add a night-specific hover colour
8	Gallery photos are 1.4–1.8 MB JPEGs	Slow gallery on mobile	Resize to about 1600 px WebP/AVIF with srcset

17 / ENGINEERING

Roadmap

Findings were intentionally **not** fixed during the refactor so behaviour could be proven identical first. They are grouped below by effort and value.

WHEN	WORK	WHY
Now (hours)	Add Open Graph and Twitter tags with a 1200×630 night-hero image · add focus-visible styles · fix white-on-cyan active icons · alt text field in the manifest · QR code on the portfolio PDF	Highest marketing and accessibility gain per hour
Next (a weekend)	Dialog semantics for the theme popup · token contrast tweaks · resize gallery photos with srcset · Person structured data · privacy-friendly analytics	Professional polish and measurement
Later	Case-study pages for TechRise and the PCB (problem, process, result, data) · downloadable one-page résumé · light-weight sitemap and canonical tags · a repeatable screenshot test in CI	Depth for reviewers; protects against regressions

Guardrails for future design changes

- Change colour only in `css/theme.css`; if a component needs a new colour, add a token.
- Keep effects decorative: no essential information may live only in a canvas.
- Every new animation needs a reduced-motion behaviour and a performance-mode behaviour.

- Keep the theme kernel tiny and blocking; everything else stays an ES module.
- After any CSS reorder, re-run the screenshot comparison: the cascade is order-sensitive (see night.css header).

// APPENDIX

Design tokens & reference

TOKEN	DAY	NIGHT
--sky-top / --sky-bottom	#5ba3e0 / #c9e8ff	#010407 / #061530
--card-bg	rgba(255,255,255,.82)	rgba(4,8,22,.85)
--card-border	rgba(255,255,255,.95)	rgba(0,210,160,.22)
--card-shadow	0 8px 32px rgba(30,80,160,.13)	0 8px 32px rgba(0,210,160,.10)
--text-primary / secondary	#0d0d0d / #374151	#eef5ff / #9ab8d8
--text-muted / --text-label	#6b7280 / #1e3a5f	#6a8aaa / #00d2a0
--nav-text / --nav-active	#0d0d0d / #1e3a5f	#ddeeff / #00d2a0
--nav-scrolled-bg	rgba(162,207,254,.90)	rgba(1,4,9,.88)
--overlay-bg	#a2cffe	#010d1e
--accent	#1e3a5f	#03fcf0
--chip-bg / --chip-border / --chip-text	rgba(255,255,255,.55) / rgba(37,99,235,.35) / #1f2937	rgba(0,210,160,.07) / rgba(0,210,160,.35) / #00d2a0
--cursor-primary / light / dark / glow	#1e3a5f / #4a7aa8 / #0f1f33 / 30,58,95	#03fcf0 / #7ffef8 / #01a89e / 3,252,240

Motion tokens

ITEM	VALUE
Theme flash	280 ms fade in · 80 ms hold · fade out · 400 ms lock tail
Popup	opacity 350 ms; first visit appears after 900 ms
Mobile menu	open 280 ms, close 250 ms, cubic-bezier(.22, 1, .36, 1); link stagger 40 ms
Day card float	6 px, 4.5–7.2 s, delay 0–2.8 s, ease-in-out, infinite
Night card canvas	24 fps · visibility margin 120 px · sky loop about 60 fps (16 ms)
Cursor	position 18% · radius 10% · effect in 3% / out 22% per frame · idle delay 1.2 s

Browser support

Evergreen browsers. Night rendering uses module Web Workers with OffscreenCanvas (Chrome 80+, Safari 15+, Firefox 114+); where unavailable, the same renderer runs on the main thread. Private fields and optional chaining require Chrome 84+, Safari 15+, Firefox 90+. No-JS visitors see the day palette and complete content.